

Owen Wells

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Education

University of California Los Angeles

Mechanical Engineering, B.S. - GPA: 3.9

Aug 2025 - Jun 2027

Mt. San Antonio College

Associate's of Science Transfer Mechanical Engineering & Mathematics - GPA: 4.0

Jul 2022 - Jun 2025

Technical Skills

CAD & Design: SolidWorks, Fusion 360, AutoCAD, Onshape, SolidWorks Assemblies, SOLIDCAM, GD&T

Analysis & Simulation: ANSYS Mechanical FEA, Structural Analysis, Simulink, MATLAB, Python

Manufacturing: Manual Mill, Manual Lathe, Drill Press, CNC Machining, 3D Printing, Rapid Prototyping, Heat Treatment

Electronics & Controls: Arduino, Raspberry Pi, ESP32, Sensor Integration, Soldering, Wiring Schematics, Circuits

Programming & Tools: MATLAB, Python, C++, Excel, Microsoft Office

Experience

Bruin Formula Racing Steering Subsystem Lead, University of California Los Angeles

May 2026 - Present

- Designed a single-U-joint steering column constrained to a 20° operating angle, balancing driver ergonomics, packaging, compliance, serviceability, and manufacturability
- Reduced steering system weight by .2 lbs, and streamlined a single U-joint steering column design for ease of assembly, serviceability and reliability at the cost of linear steering.
- Analyzed Anti-Ackermann steering geometry with the vehicle dynamics lead, forecasting a 3% increase in lateral acceleration relative to the previous parallel-steering baseline.
- Collaborated with suspension and vehicle dynamics subteams to evaluate steering geometry tradeoffs, packaging constraints, and driver-interface requirements.

Bruin Formula Racing New Member FSAE, University of California Los Angeles

Sept 2025 - April 2026

- Manufactured 30+ suspension components using manual mill and lathe operations to support FSAE vehicle assembly.
- Designed welding tab jigs in SolidWorks to improve suspension manufacturing repeatability and assembly accuracy.
- Developed steering subsystem assembly drawings, exploded views, bill of materials, and cost report to support documentation, manufacturing.

Embedded Tutor / Drop In Tutor, Mt. SAC ASAC - Walnut, CA

Aug 2024 - Jun 2025

- Tutored students in physics, engineering, strengthening technical communication and analytical problem-solving skills.
- Led group tutoring sessions by facilitating collaborative activities and adapting explanations to different learning styles.

Projects

Carbon Fiber A-Arms, Bruin Formula Racing

Oct 2025 – Present

- Designed carbon fiber A-arms for manufacturability and assembly with a target 70% weight reduction.
- Optimized tube diameter and length geometry for a 406 lb compressive load case, reducing excess material while maintaining an acceptable margin of safety.
- Iterated aluminum wafer geometry using ANSYS Mechanical FEA to identify stress concentrations, reduce mass, and evaluate load-bearing performance resulting in a maximum acceptable stress of 62,216 psi at the load-bearing region.

Electronics & CAD Lead, Spy Watch, 2025 UCLA Engineering HACK Competition

Jul 2025 - Aug 2025

- Programmed a Raspberry Pi Pico to acquire sensor data, transmit readings to a web interface, and display real-time system feedback through an LCD.
- Designed compact enclosure geometry to optimize component packaging, sensor accessibility, and wiring layout.
- Integrated a DC voltage regulator circuit to step a 3.7 V lithium-ion battery output to a regulated 5 V, improving ESP32-CAM reliability during Wi-Fi current spikes.

Water Level Indicator & Regulator Breadboard Project

Oct 2024 - Dec 2024

- Designed and built a breadboard-based monitoring and control circuit to detect three discrete water levels and three temperature ranges with comparator logic, driving LED indicators and automated pump control to regulate water-level.
- Implemented a thermistor Wheatstone bridge, 2 V/V difference amplifier, and comparator network to trigger temperature indicators at defined voltage thresholds.
- Designed an RC-delay pump shutoff circuit using an NPN power transistor, enabling 10 s of controlled drainage based on a pump flow rate of 30 mL/s.